Hot Engines and fire prevention

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***Abstract*— Skin that is left in the summer sun does not need a flame to burn; enough heat alone will do it. The engine surface behaves the same way. Once it reaches a certain temperature, the fuel that makes contact can ignite, without a spark or open flame. The work presented here has been undertaken to understand how heat transfer governs that ignition and how the surface temperature behind it can be predicted and controlled before ignition occurs. The three modes of heat transfer, conduction, convection, and radiation across an operating engine have been studied together to locate the regions where surface temperatures reach the autoignition limits of basic flammable fluids. The significance of this work comes from a change in how hazard is treated. Most work has relied on sensing the fire and then suppressing it after ignition, which accepts some damage as a starting point. The approach taken here treats the hot surface as the variable to be controlled, so that the path to ignition is broken before a flame is formed. Earlier engine fire safety has largely depended on fixed insulation, manual inspection, and suppression after the fact, each of which responds only after ignition. The methodology developed here has instead been built around predictive modeling of where surfaces become dangerously hot, so that high-risk zones can be discovered during the design process rather than on the road. A comparison with these earlier attempts has shown that predictive thermal mapping provides earlier warning and finer spatial detail than either inspection or suppression alone. It is expected that the completed work will produce a validated framework linking engine operating conditions to surface temperature fields, and from those fields to a measurable margin against ignition. Such a framework has value as it turns fire prevention into a design parameter that can be specified and tested before any fire damage has taken place. The results are expected to guide safer engine housing and better placement of thermal barriers and cooling paths. The same reasoning can be expected to carry across automotive, aerospace, and power industries, wherever hot surfaces and flammable fluids share a confined space.**

Keywords*— *hot-surface ignition, thermal management, engine fire prevention, surface temperature prediction, autoignition, heat transfer analysis, predictive thermal design.

I. INTRODUCTION

Engine compartments inherently combine highly flammable fluids with dangerously hot surfaces, creating a persistent and severe fire hazard across the automotive, aerospace, and power generation industries. Traditionally, fire safety within these confined environments has heavily relied on reactive measures, such as manual inspections, fixed insulation, and chemical suppression systems that deploy only after ignition has already occurred. Because these conventional methods fundamentally rely on sensing an active fire, they accept a certain degree of initial structural and component damage as an unavoidable starting point. However, fire prevention can be approached more proactively by treating the engine's hot surfaces as controllable variables rather than static, inevitable hazards. When a flammable fluid makes contact with a surface that has exceeded its specific autoignition temperature, ignition can occur entirely without the presence of an open flame or electrical spark. Therefore, understanding the heat transfer mechanisms that govern this surface heating is critical to breaking the path to ignition before a flame can ever form.

This paper introduces a predictive methodology built around the thermal mapping of engine components to identify high-risk, hot-surface zones during the mechanical design phase. By analyzing the combined effects of conduction, convection, and radiation across an operating engine, this study locates the specific regions where surface temperatures are likely to reach the autoignition limits of common flammable fluids. Ultimately, this work seeks to establish a validated predictive framework that links engine operating conditions directly to surface temperature fields. By transforming fire prevention into a measurable design parameter, this approach provides earlier warnings and finer spatial detail than traditional methods, allowing engineers to optimize engine housing and thermal barriers long before the system is deployed.

II. LITERTURE REVIEW

A. Hot-Surface Ignition and Surface Temperature Prediction

This category gathers the work that treats the hot surface itself as the thing to understand and predict, because fuel will not ignite on a surface that stays below its autoignition temperature. The studies below move from measuring temperatures, through fitting probability models, to simulating ignition before it occurs.

The ignition of lubricating oil on hot surfaces has been studied on a purpose-built test rig that paired high-speed imaging with statistical analysis [1]. The lowest temperature at which the oil ignited was 311 degrees Celsius, and once the surface passed roughly 407 degrees, the chance of ignition rose sharply. From the recorded outcomes, a probability model was fitted that estimates how likely ignition occurs at a given surface temperature. This work also separated combustion into distinct stages and tracked ignition delay, flame duration, and flame height. For this project, these values link a measurable surface temperature directly to a quantified ignition risk.

A second experiment focused on automotive lubricating oil placed on a heated plate in controlled volumes [2]. The ignition temperature, flame height, flame propagation speed, and flame temperature were all measured as the surface temperature increased. A useful takeaway is how the ignition core sat above the plate near 370 to 390 degrees at 5.5 to 6 centimeters, then dropped to 3 to 4 centimeters as the surface approached 480 degrees. This result shows that surface temperature shapes not only whether the oil ignites but where the flame first anchors, which is the kind of information a thermal map of an engine compartment could use during design, to keep surfaces below their autoignition temperature for as long as possible.

Prediction does not have to be experimental. A computational study modeled pre-ignition caused by lubricating oil droplets inside the combustion section of a natural gas engine [3]. The simulation used a CFD code with a mesh built around the real engine structure, a stripping model that decided when oil left the piston ring, and a model that tracked the thermal and chemical state of each droplet. Resolving the heat exchange between a hot droplet and the surrounding gas lets the study reproduce the pre-ignition mechanism numerically. It points toward predicting ignition from simulated temperature fields rather than waiting to measure them.

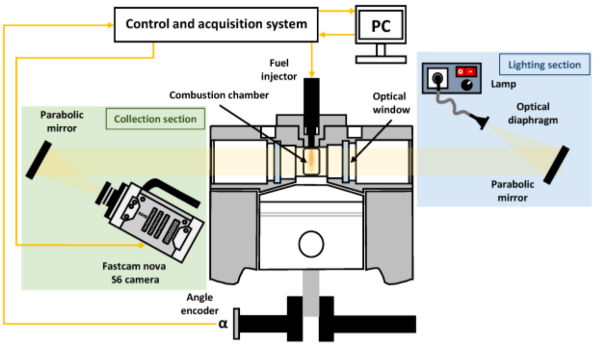

Figure 1. CFD Modeling approach to fuel droplets inside a combustion chamber [3].

Why fuel ignites on hot surfaces can also be traced back to its thermal decomposition [4]. Thermogravimetric analysis was paired with hot-surface tests on several common fuels, and ignition occurred as a probability instead of at one exact temperature, so it could not be fixed to a single ignition temperature. The lowest ignition temperatures, measured at a 5 percent probability, fell inside the range where the fuel was already giving off combustible gas as it evaporated. A model was built for how the vapor concentration and temperature vary above the surface. The study shows that predicting ignition means predicting the gas released when heated, not just the surface reading.

A similar hazard in cars was approached through image analysis [5]. Lubricating oil leaking onto a heated surface was filmed, then processed with edge detection and a fractal-dimension method to describe the flame outline. Ignition delay shortened as the surface heated, with 450 degrees marked as a turning point. A possible-ignition zone was found between the overlap of clear ignition and clear non-ignition, whose upper and lower boundaries were fitted with equations carrying R-squared values above 95 percent. The fractal dimension then tracked how the flame outline stabilized over time. These boundaries give an usable rules for flagging risky surface temperatures.

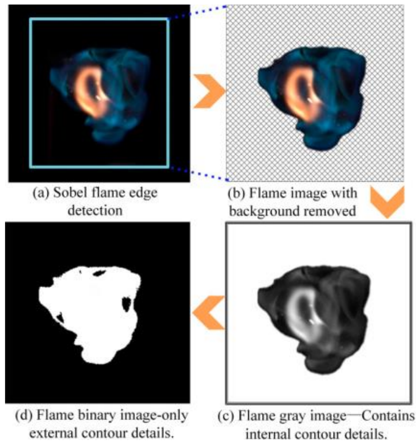

Figure 2. Flame image analysis [5].

Aviation fuel has been tested the same way, because jet kerosene leaking near a hot surface carries the same risk in the air [6]. Leaked RP-3 kerosene was dropped onto a heated surface and filmed with a high-speed camera, and the critical temperature at which ignition reached a 50 percent probability was found by statistics. From the evaporating fuel, a vapor plume model was built for the minimum surface temperature that would ignite a leak. The measured temperature and vapor concentration above the surface changed continuously with time. The model gives a quantitative basis for judging when a hot aircraft or engine surface becomes an ignition source.

Table 1. Hot-Surface Ignition and Surface-Temperature Prediction

Ref	Approach	Result	Pros	Cons
[1]	Experimental test rig, high-speed imaging, statistical analysis	Lowest ignition at 311°C; probability rises sharply above 407°C; ignition-probability model fitted	Direct quantified link between surface temperature and ignition risk; tracks distinct combustion stages	Limited to one oil type; lab rig, not a real engine environment
[2]	Heated-plate experiment, varied oil volume	Ignition core height falls from about 6 cm to 3 to 4 cm as the surface nears 480°C	Adds spatial detail on where the flame anchors; open access with figures	Flat plate, not a curved or real engine surface; oil volume range may not match field conditions

[3]	CFD simulation, droplet stripping and single-particle ignition-cell models	Reproduced lube-oil-drop pre-ignition mechanism numerically inside a real engine geometry	Predicts ignition without physical testing; built on actual engine grid	High computational cost; needs experimental validation; paywalled, no public figures
[4]	TG-FTIR paired with hot-surface tests, several common liquid fuels	Ignition is probabilistic, not fixed; lowest ignition temperatures align with combustible-gas generation range; vapor model built	Connects ignition to the chemistry behind it, not just surface reading	Tested on lab fuels, not directly on engine-representative surfaces
[5]	Image analysis, edge detection, fractal-dimension method, automotive	450°C marked as a turning point; possible-ignition zone boundaries fit with R^2 above 95%	Gives a practical boundary rule for flagging risk; open access	Fractal-dimension metric is less intuitive for direct design use
[6]	Experimental, high-speed imaging, RP-3 aviation kerosene on a hot wall	Critical temperature for 50% ignition probability found; vapor plume model for minimum ignition surface temperature	Extends the hazard to the aerospace case; quantitative prediction model; open access	Conference paper, lighter on data than a full journal study

B. Bridging Simulation and Reality Through Strict Design Tolerances

To effectively prevent engine fires through predictive thermal mapping, the physical manufacturing of engine components must strictly adhere to their original design specifications. A predictive model is only as reliable as the physical system it represents. For example, applying a thermal optimization design method to aluminum alloy pistons in diesel engines relies on precise dimensional and material specifications to properly manage heat distribution [7]. If these optimized specifications are not strictly maintained during manufacturing, the base metal may fail to dissipate heat effectively, leading to localized hot spots that pose a severe ignition risk.

This strict adherence to design tolerances is equally critical for the long-term integrity of thermal barrier coatings (TBCs) applied to these base components. The performance of TBCs inside internal combustion engines relies on exact material formulations to resist severe thermo-chemical degradation over time. Assessments of corrosion resistance in these coatings demonstrate that any deviation from the specified microstructural integrity allows corrosive combustion byproducts to prematurely degrade the barrier **[8].** When manufacturing variations compromise this resistance, the coating spalls, unexpectedly exposing the underlying metal to extreme temperatures and creating a sudden hot-surface ignition hazard.

Maintaining these precise surface temperatures is crucial because the flammable fluids in the engine bay are highly sensitive to even minor thermal variations. Experimental shock tube studies analyzing the ignition of various lubricating oil compositions reveal that ignition delay times and autoignition thresholds are strictly governed by specific localized temperature and pressure states [9]. If manufacturing defects in engine geometry or cooling channels create unpredicted thermal hotspots, it can force the surrounding environment to cross these exact ignition thresholds, instantly turning a minor oil leak into a catastrophic fire.

To map and prevent these underhood hotspots before they occur, engineers rely heavily on predictive computational frameworks. For instance, evaluating the underhood thermal management of heavy machinery utilizing 1D-network simulations allows designers to accurately predict component temperatures and cooling airflows during the design phase **[10]**. However, this predictive accuracy completely collapses if the physical specifications of the manufactured parts such as heat exchanger dimensions or fan clearances deviate from the digital 1D network, allowing unchecked heat to accumulate near volatile fluids.

Similar geometric precision is mandatory for the active components responsible for cooling those exact flammable fluids. Structural optimizations of vehicular oil coolers based on advanced 3D impermeable flow models demonstrate that heat dissipation is highly dependent on the exact flow path and structural design of the cooler **[11]**. If manufacturing flaws alter these highly optimized internal geometries, the cooler will fail to properly shed heat, leaving the lubricating oil dangerously hot and significantly increasing the likelihood of hot-surface ignition upon leakage.

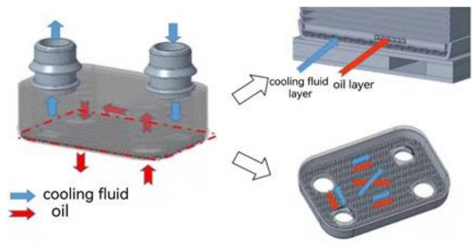

Figure 3. Water side fin structure (left) and oil side fin structure (right) [11].

Finally, the physical placement and structural integrity of passive thermal management components are crucial for mitigating risks when active cooling shuts down entirely. Investigations using underhood test rigs demonstrate that parts near hot manifolds continue to climb in temperature for several minutes after the engine is turned off due to severe heat soak [12]. Maintaining the specified dimensions and exact mounting locations of heat shields is critical, as they successfully block this radiated heat during the vulnerable soak period, keeping nearby surfaces well below the autoignition threshold of leaking fluids.

Table 2. Design Tolerance Impacts on Thermal Validation

Ref	Approach	Result	Pros	Cons
[7]	Thermal optimization design method applied to aluminum alloy diesel pistons.	Precise material and geometric specifications drastically reduce peak surface temperatures.	Proactive design minimizes localized hot spots and prevents base metal failure.	Highly sensitive to manufacturing deviations; unoptimized parts pose severe ignition risks.
[8]	Assessment of corrosion resistance of TBCs on internal combustion engine components.	Confirms coating longevity depends heavily on resisting thermo-chemical degradation.	Highlights the real-world operational durability needed beyond just thermal resistance.	Coatings failing to meet anti-corrosion specs will rapidly spall and expose base metals.
[9]	Shock tube experiments analyzing the ignition of various lubricating oil compositions.	Ignition delay and autoignition are strictly governed by precise localized temperatures.	Provides exact threshold data for the autoignition limits of common engine fluids.	Highly sensitive; unpredicted hot spots from poor tolerances easily trigger these exact thresholds.
[10]	1D-network simulations for predictive underhood thermal management.	Accurately predicts component temperatures and cooling airflows proactively.	Enables rapid, proactive thermal mapping without expensive physical prototypes.	Predictive validity is completely voided if physical clearances and dimensions deviate from the model.
[11]	3D impermeable flow modeling for structural optimization of vehicular oil coolers.	Proper heat dissipation relies entirely on highly specific optimized internal geometries.	Maximizes the cooling efficiency of the volatile lubricating oil.	Minor manufacturing defects disrupt the calculated flow, leaving the oil dangerously hot.
[12]	Simplified underhood test rig tracking temperatures during a 100-minute soak after shutdown.	Unshielded parts near hot manifolds continue to climb in temperature due to heat soak.	Confirms the danger window post-shutdown and the effectiveness of heat shields.	Relies entirely on the strict physical placement and structural integrity of the passive shields.

C. Thermal Barriers and Surface Cooling for Ignition Prevention

Part A was all about how hot a surface needs to get before fuel sitting on it actually lights up, and this part flips that around and looks at how you keep the surface from getting that hot in the first place. The six papers referenced below in part C are split pretty evenly between two approaches, [13] and [17]. A 2023 study ran a finite element model on a diesel piston, lining up a bare aluminum piston against the same piston wearing a thermal barrier coating, then ran a multi objective optimization to pick the best mix of thickness and material [13]. Peak temperature on the coated piston dropped 55.54 degrees Celsius compared to the bare one, about a 15 percent cut, and thicker ceramic kept giving better insulation, although it drags more stress along with it the thicker you go. Fifty five degrees is a sizeable number when the conversation is about margin against ignition, and it is coming from an actual measurement, not a simulation guess.

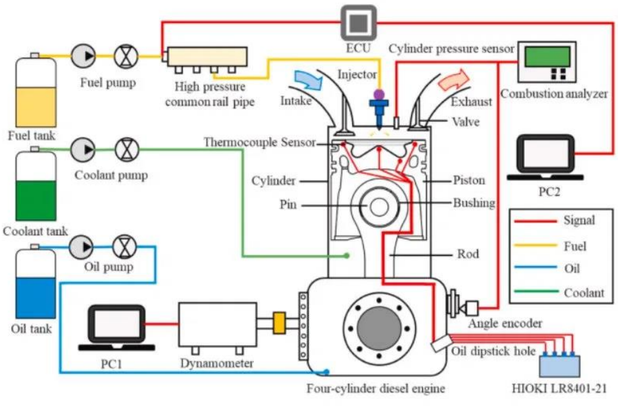

Figure 4. Experimental diesel engine test setup with fuel, coolant, and oil systems and combustion instrumentation. Image by Liu et al. [13].

Pistons are not the only place this gets tested though. Researchers ran low thermal inertia thermal barrier coatings through a full 100 hour durability cycle on spark ignition engine pistons, throwing in cold start comparisons against an uncoated baseline the whole way through [14]. Nothing flaked or failed across the entire run, and the coated pistons also cut particulate matter and unburned hydrocarbons during the cold starts. A coating that cannot survive its own durability test is not worth much for fire safety either, so the fact this one held up matters just as much as the emissions numbers do.

A coating can look great sitting at idle and then behave completely differently once the engine is actually under load, which is why one team optimized their coating's performance across an entire drive cycle instead of testing at a single fixed point [15]. Their thermo mechanical model tracked how the coating held up as conditions kept shifting through a simulated drive. A single ignition margin number pulled from one test point does not say much on its own, a surface running safely cool while cruising could act completely different under hard acceleration, so any real thermal map needs to follow the engine through its whole duty cycle rather than one snapshot of it.

Turbine blades get the same coating treatment engines do, and one study went after how much a coating's thickness and surface roughness actually swing its cooling performance [16]. Both turned out to matter quite a bit. That tells you a coating is not some fixed spec you slap down once and walk away from, the exact finish and thickness that ends up on the part can shift the result, and that same kind of sensitivity is probably sitting inside engine bay coatings too, where manufacturing variation could end up deciding whether a surface stays cool or drifts toward an ignition risk.

Cooling does not have to mean one strategy for the whole engine either. A team built an AI controlled dual loop coolant system that ran two separate loops instead of one, aiming to cut fuel use while still holding engine temperature where it needed to be [17]. Splitting the loops let different parts of the engine get treated differently instead of forcing one blanket cooling approach onto everything, which fits how a thermal map for an engine bay should probably work, since the manifold and turbo run far hotter than the parts sitting further out and likely need their own cooling logic instead of sharing one with everything else.

What happens once the airflow shuts off entirely is its own problem. A test rig study built two simplified engine bays, a car layout and a truck layout, heated both up to realistic running temperatures, then cut the fan and heaters and just watched what happened over a 100 minute soak [18]. Parts near the hot manifold kept climbing for the first ten minutes after shutdown before they started cooling, driven by radiation taking over once forced convection stopped, and the layout with a heat shield between the manifold and nearby cold parts stayed noticeably cooler the whole time compared to the layout with no shield at all. That lines up pretty directly with the ignition window from Part A, because the riskiest moment for a fuel leak is not necessarily mid drive, it can be right after shutdown, when the air stops moving but the nearby parts are still soaking up leftover radiated heat.

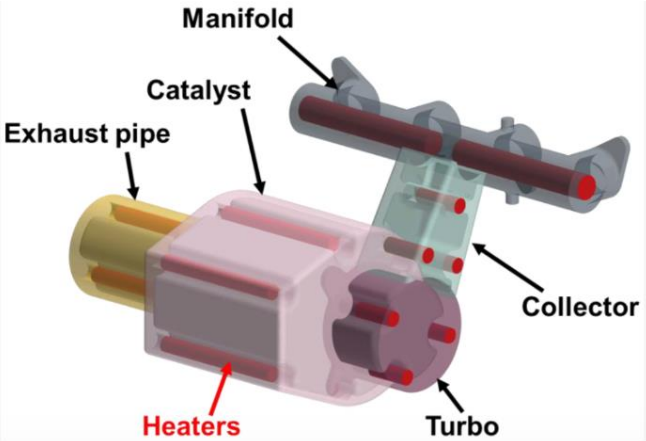

Figure 5. Hot engine components and heater positions in the underhood soak test rig. Image by Vdovin and Kadri Sathiyan [18].

Table 3. THERMAL BARRIER AND COOLING APPROACHES

REF	APPROACH	RESULTS	COMMENTS
13	TBC THICKNESS AND MATERIAL OPTIMIZED VIA A MULTI OBJECTIVE ALGORITHM ON A DIESEL PISTON	PEAK SURFACE TEMPERATURE REDUCED BY 55.54°C (15.08%) VERSUS THE BARE ALUMINUM PISTON	STRONG, DIRECTLY MEASURED MARGIN GAIN, THOUGH THICKER COATINGS RAISE THERMAL STRESS TRADEOFFS
14	LOW THERMAL INERTIA TBC TESTED ON SPARK IGNITION ENGINE PISTONS THROUGH A 100 HR DURABILITY RUN	COATING SURVIVED THE FULL DURABILITY TEST AND CUT PARTICULATE AND UNBURNED HYDROCARBON EMISSIONS AT COLD START	SHOWS A BARRIER NEEDS TO SURVIVE REAL OPERATING LIFE, NOT JUST LAB CONDITIONS
15	TBC PERFORMANCE OPTIMIZED ACROSS A FULL DRIVE CYCLE INSTEAD OF ONE FIXED OPERATING POINT	IDENTIFIED THAT COATING BEHAVIOR SHIFTS SIGNIFICANTLY WITH CHANGING LOAD AND SPEED	A SINGLE POINT MARGIN NUMBER IS NOT ENOUGH; DESIGN MUST COVER THE FULL DUTY CYCLE
16	STUDIED THE EFFECT OF TBC THICKNESS AND SURFACE ROUGHNESS ON TURBINE	BOTH VARIABLES SIGNIFICANTLY CHANGED COOLING PERFORMANCE SENSITIVITY	MANUFACTURING VARIATION COULD UNDERCUT A BARRIER'S REAL WORLD PROTECTION

	BLADE COOLING		
17	AI CONTROLLED DUAL LOOP COOLANT SYSTEM TARGETING DIFFERENT ENGINE ZONES SEPARATELY	REDUCED FUEL CONSUMPTION WHILE MAINTAINING TARGET TEMPERATURES ACROSS BOTH LOOPS	SUGGESTS ZONE SPECIFIC COOLING COULD BE SMARTER THAN ONE BLANKET STRATEGY
18	SIMPLIFIED TEST RIG MEASURED PART TEMPERATURES THROUGH HEAT UP AND A 100 MIN SOAK, WITH AND WITHOUT A HEAT SHIELD	SHIELDED PARTS RAN COOLER DURING SOAK; UNSHIELDED PARTS KEPT RISING FOR ~10 MIN AFTER SHUTDOWN	CONFIRMS THE DANGER WINDOW INCLUDES THE PERIOD RIGHT AFTER ENGINE SHUTOFF, NOT JUST RUNNING

D. Abbreviations and Acronyms

CFD — Computational Fluid Dynamics

RP-3 — designation for a common aviation kerosene fuel

TG-FTIR — Thermogravimetric Analysis and Fourier Transform Infrared Spectroscopy

TBC - Thermal Barrier Coatings

HT- The High-Temperature Loop

LT- The Low-Temperature Loop

WCAC - Water-Cooled Charge Air Cooler

WCOND - Water Condenser.

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